

Antecedents and consequences of team sensemaking capability in product development projects

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With the increasing popularity of organizational sensemaking in the literature, sensemaking capability of firms attracts many researchers and practitioners from different fields. Nevertheless, sensemaking capability is rarely addressed in the new product development (NPD) project teams in the technology and innovation management literature. Specifically, we know little about what team sensemaking capability is, its ingredients and benefits, and how it works in NPD projects (e.g., its antecedents and consequences). By investigating 92 NPD project teams, we found that (1) team sensemaking capability, which is composed of internal and external communication, information gathering, information classification, building shared mental models, and taking experimental actions, has a positive impact on the information implementation and speed-to-market; (2) information implementation and speed-to-market mediate the relationship between team sensemaking capability and new product success; and (3) team sensemaking capability mediates the relationship between team processes and information implementation and partially mediates the relationship between team processes and speed-to-market. We also found that team autonomy, interpersonal trust among team members, and open-mindedness of team members positively influence the development of team sensemaking capability. Theoretical and managerial implications of the study findings are discussed.

1. Introduction

The concept of sensemaking in teams is studied from different perspectives and within different contexts in the literature such as technology adaption (Bansler and Havn, 2006; Seligman, 2006), new product development (NPD) (Akgün et al., 2006b),

health care (Albolino et al., 2007; Davidson, 2010; Kitzmiller et al., 2010; Manojlovich, 2010), and battle fields (Jensen, 2009). In those studies, researchers indicate that sensemaking is an ongoing process of meaning construction, which reflects how team members make interpretations of actions and events, thus making ambiguous situations more orderly, and

highlights how they develop a common or shared understanding (Christiansen and Varnes, 2009). In this way, team sensemaking is perceived as an important factor for teams' operations and performances. For instance, Klein et al. (2010) note that most failures can be traced to a breakdown in team sensemaking where critical cues were ignored and the teams failed to synthesize the existing information.

Interestingly, it should be noted that while most of the studies focus on the sensemaking concept as a portrait of the 'organizing' in the team context (Ashmos and Nathan, 2002; Klein et al., 2010), a few studies mention about it as a specific *capability* (Weick et al., 2005). Awazu (2004), for instance, notes that sensemaking is an important capability for organizations and groups because it integrates people's thoughts and generates new knowledge as well as constituting an ability to reconfigure existent knowledge to solve current problems. Nevertheless, we know less about how to conceptualize and then operationalize the team sensemaking capability in the technology innovation management (TIM) literature. Also, from a managerial point of view, the antecedents and consequences of sensemaking capability in teams warrant an empirical investigation.

In order to deal with the issues mentioned earlier, based on the studies of Weick (1995), Weick et al. (2005), and Neill et al. (2007), we put forward that team sensemaking capability can be conceptualized as a compendium of interrelated routines and practices about 'what' the new technology and market- and project-related information mean, 'how' it is assimilated and interpreted, and 'which' actions are considered for a better product development process in teams. In a sense, team sensemaking capability indicates the practices by which team members (1) interpret the technological-, market-, and project-related information/knowledge through interaction with others; (2) construct their cognitive accounts that allow them to comprehend the product development project as a whole; and (3) respond to market-, technology-, and product-related events collectively. Also, regarding the operationalization of team sensemaking capability, based on the generic organizational/team sensemaking literature, we argue that team sensemaking capability is a second-order construct composed of internal/external communication (Weick et al., 2005; Neill et al., 2007), information gathering (Thomas et al., 1993; Neill et al., 2007), information classification (Akgün et al., 2006b), building shared mental models (Bogner and Barr, 2000; Neill et al., 2007), and taking experimental actions (Bogner and Barr, 2000; Weick et al., 2005). Next, for the antecedents and consequences of team sensemaking capability, based on the study of

Akgün et al. (2005), we used the components of team processes, such as trust between team members, team member familiarity, and team member proximity, as determinants. As team sensemaking capability is a social act that is grounded both in individual activity and social interactions (Weick, 1995; Maitlis, 2005), it is necessary to pay attention to psychological processes and interrelations between team members that enable sensemaking (Reuben, 2007). Indeed, sensemaking capability is a communicative practice in which people interpret their environment and the surroundings of the project, technology/market-related information, as well as the stimuli through the interaction with others. Also, as team sensemaking capability requires flow of information to bring out meaning and obtain plausible interpretations (Daft and Weick, 1984), trust, familiarity, and proximity enable team members to capture the information they need. In addition to the components of interpersonal relationships, based on the studies of Ashmos and Nathan (2002) and Neill et al. (2007), we also incorporated the autonomy and open-mindedness variables as the determinants of team sensemaking capability. In fact, Neill et al. (2007) reveal the positive relationship between open-mindedness and sensemaking capability at the organizational level, which warrants an empirical investigation at the team level. As team sensemaking capability requires the changing of the existing thought or behavior patterns by exploring different ways of thinking and doing things (Ashmos and Nathan, 2002), it unveils the importance of open-mindedness that enables teams to question long-held practices and beliefs (Sinkula et al., 1997). Regarding the autonomy variable, Ashmos and Nathan (2002) suggest an empirical study for their advice to teams that they should increase their autonomy to be more successful at sensemaking. Finally, consistent with TIM literature, as the consequences of team sensemaking capability, we investigate speed-to-market, information implementation,¹ and new product success to test the team sensemaking capability/project outcome link. For instance, Langerak and Hultink (2006) suggest that firms that succeed to introduce new products to the market faster than their competitors can obtain first-mover advantages. Chen et al. (2005) also ascertain that 'the greater speed-to-market, the greater the competitive advantage over later entrants'. Information implementation (i.e., detection and correction of errors during the NPD process) is also identified as an outcome in this study because knowledge is a source of sustainable competitive advantage (Atuahene-Gima and Murray, 2004). The reason why we choose information implementation as the other consequence of team sensemaking capability is the

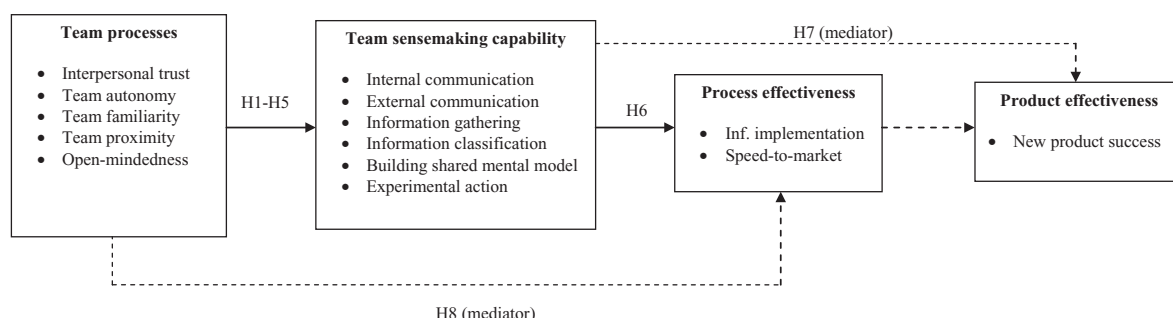


Figure 1. Proposed model.
Note: Dotted line denotes the mediator hypothesis.

belief that traditional emphasis on meeting time and budget constraints is no longer sufficient to guarantee the achievement of the project objectives (Patanakul and Shenhar, 2011). Instead, project teams must be able to utilize information better and correct the product-related problems to achieve the outperforming business results in market places. With regard to new product success, Olson et al. (1995), for instance, propose that the most critical performance issues must typically identify whether the resulting product meets market, technological, and managerial expectations. In this study, we classified information implementation and speed-to-market as process effectiveness, and new product success as product effectiveness variable, based on the studies of scholars (e.g., Brown and Eisenhardt, 1995; Moorman and Miner, 1998).

Therefore, as shown in Figure 1, this study investigates (1) the impact of team sensemaking capability on the project process effectiveness variables (e.g., information implementation and speed-to-market); (2) the mediating role of process effectiveness between team sensemaking capability and product effectiveness (new product performance in the market place); (3) which team process characteristics (e.g., trust, familiarity, proximity, autonomy, and open-mindedness) improve team sensemaking capability; and (4) the mediating role of team sensemaking capability between process effectiveness and team processes.

2. Team sensemaking capability

Sensemaking capability is the team's ability to interpret, understand, and construct meaning on the project-, technology-, and market-related information, and then act based on that information to complete the project successfully² (Gioia and Chittipeddi, 1991). Especially, NPD teams have a better understanding of the uncertainties they face during the

product development processes by developing sensemaking capability (Ashmos and Nathan, 2002). Sensemaking capability, for instance, helps teams to identify events that recur in the environment and make them more predictable (Weick, 1995), and structure the information gathered to find out meaningful patterns in customer tendencies, competitor strategies, and sectorial trends. Also, team sensemaking capability provides a body of knowledge about project-related activities, such that sensemaking capability helps teams to share the summarized information with all team members and combine customer-competitor information, tacit knowledge, technological competency, insights, and ideas into the new product concept (Dougherty et al., 2000; Akgün et al., 2006b). For example, Mohrman et al. (2003, p. 9), in this respect, wrote that 'Several engineers may examine surprising test results and share their different disciplinary perspectives and experience-based tacit understandings about why these results occurred. This form of collective sensemaking, entailing personal links between people with different knowledge, can yield new knowledge. Knowledge can be linked across the organization if it can be accessed through interactions, or transactions'. Next, team sensemaking capability develops a collective belief and constructs shared mental models, shaping what people perceive as technological, marketing, and organizational realities. In a sense, sensemaking capability is instrumental in shaping product development process regarding the nature of new information, rendering it more comprehensible, meaningful, or desirable, and therefore more legitimate.

Based on the sensemaking theory structured on the three-step perspective (noticing, interpretation, and actioning)³ (see Weick, 1995; Weick et al., 2005), we further argue that, from an operational perspective, team sensemaking capability has five subdimensions in the NPD context including the internal/external communication (Weick et al., 2005; Neill et al., 2007), information gathering (Daft and Weick, 1984;

Thomas et al., 1993; Neill et al., 2007), information classification (Akgün et al., 2006b), building shared mental models (Bogner and Barr, 2000; Neill et al., 2007), and taking experimental actions (Bogner and Barr, 2000; Weick et al., 2005). Specifically, internal communication is the vehicle through which team members from different functional areas share information (Pinto and Pinto, 1990). For instance, Weick et al. (2005) suggest that 'sensemaking is, importantly, an issue of language and communication. Situations, organizations and environments are talked into existence'. Indeed, sensemaking capability requires talking, interaction, conversation, argument, and dialogue with others where teams offer a platform to talk, argue, and interpret (Ashmos and Nathan, 2002). In this way, team members reduce equivocality through social negotiation of interpretations and develop an understanding of their environment (Neill et al., 2007).

External communication shows the degree of contact with outsiders, such as customers and suppliers (Brown and Eisenhardt, 1995). Team sensemaking capability has an external focus as it depends on the availability of information from the environment. For instance, each team member accesses not only self-contained information from his or her functional background as intended but also knowledge from diverse external personal networks (Edmondson and Nembhard, 2009).

Information gathering indicates the collection of customer- and competitor-related information by salespeople and market research personnel, forming a customer council, meeting with users, creating mechanisms for integrating customer requirements into product design, etc. (Neill et al., 2007). Here, gathering environmental information provides a raw material for sensemaking capability, and collected information is progressively assimilated and focused on team actions.

Information classification includes labeling, indexing, sorting, abstracting, and coding information (Akgün et al., 2006b). In a team setting, organizing or coding information into meaningful clusters impacts the team effectiveness and the rate at which teams make sense (Akgün et al., 2006b), and allows team members to verify findings by comparing similar cases, the past and the present and to answer task-specific questions. Also, when all information is broken down into separate categories that concentrate just on a single area, extraneous 'noise' is filtered out.

Shared mental models are knowledge structures and shared understandings held by team members that enable them to form accurate explanations and expectations for the task, and in turn, to coordinate their actions during the NPD process (Dougherty

et al., 2000). As Neill et al. (2007) suggest for organizations, through these shared mental models, NPD teams make sense of the situation by encoding and assigning meaning to environmental cues. Shared mental models also help the teams distill the voluminous data and perceptions they gather from a complex and turbulent market environment in a more manageable set of cognitive categories.

As the last component of the sensemaking capability, experimental action is an improvisational process to test customers' reactions to a new product's prototype on a trial basis (Jeong and Brower, 2008). The literature indicates that although interpretation is a component of sensemaking, actual sensemaking can only occur as action is taken. For instance, under conditions of ambiguity and uncertainty, teams are counseled to undertake experimental actions in an attempt to collect information about the environment (Bogner and Barr, 2000). Bogner and Barr (2000, p. 220), for instance, wrote that 'These actions are consistent with action-based sensemaking in that they are undertaken with little knowledge *ex ante* about whether the action will be successful. Thus, while the intent is to gain advantage, the actions are experimental'. Brown and Eisenhardt (1995) also reveal that product teams who engage in more experiential or improvisational product design through frequent iterations develop products more quickly, such that actions create the necessary conditions for further action (Weick, 1995) and produce an essential part of the environment in which they occur.

Having established the characteristics of team sensemaking capability, we now develop arguments regarding the role of team processes on the team sensemaking capability and how team sensemaking capability impacts a project's and a product's success.

3. Hypothesis development

3.1. Antecedents of team sensemaking capacity

In this study, we investigate the interpersonal trust, team autonomy, team member familiarity and proximity, and open-mindedness as the antecedents or drivers of the team sensemaking capability in NPD projects.

Interpersonal trust is the extent to which a person is confident in and willing to act on the basis of the words, actions, and decisions of another (McAllister, 1995). We discuss that interpersonal trust fosters sensemaking capability by dissolving psychological barriers that prevent both explicit and tacit knowledge sharing between team members, and by improv-

ing team members' agreement on the professional judgments of each other, all of which work in different functions, without being able to fully understand how those decisions were made (Madhavan and Grover, 1998; Barczak and McDonough, 2003). Interpersonal trust also strengthens the combination of information/knowledge and perspective exchange between team members (Brown and Eisenhardt, 1995) by helping to anticipate new events and act collectively (Klein et al., 2010). Interpersonal trust further enables team members to question long-held practices and mental models, as they feel confident in discussing any topic related to NPD without the fear of any embarrassment or punishment – enhancing sensemaking capability. Therefore:

H1: Interpersonal trust is positively related to NPD team sensemaking capability.

Team autonomy indicates the extent to which the team is responsible for its own work activities (Gerwin and Moffat, 1997). An effective sensemaking capability necessitates the NPD team members to be autonomous. In fact, sensemaking capability breeds a questioning process of organizational routines and processes (Klein et al., 2010), which can only be achieved by autonomy. For example, autonomy helps team members to keep the team from performing outlived applications that lost their usefulness as they have enough opportunity and authority for free thinking (Ashmos and Nathan, 2002). Also, sensemaking capability necessitates fast decisions and quick actions that can only be managed by autonomous teams that have the authority and responsibility to make fast decisions based on the real-time information (Moorman and Miner, 1998). Therefore:

H2: Team autonomy is positively related to NPD team sensemaking capability.

Team member familiarity refers to interactions between the group members (Harrison et al., 2003). Familiar team members possess more knowledge about each others' skills, perspectives, and interpersonal styles, which can facilitate coordination of efforts and improve the use of specialized information as each one of them is aware of who knows what (Akgün et al., 2006a). This high-level involvement in information sharing fosters sensemaking capability by establishing shared mental models that help team members to develop cognitive maps of their environment (Weick, 1995). In addition, familiar people feel more comfortable about expressing their disagreement, alternative perspectives, and judgments that improve the sensemaking capability (Ashmos and Nathan, 2002). Further, as prior interactions between team members reduce uncertainty

and anxiety about social acceptance during the NPD project, and promote interpersonal attraction and cohesiveness, it results in higher level of information sharing between team members, which facilitates sensemaking capability. Therefore:

H3: Team member familiarity is positively related to NPD team sensemaking capability.

Team member proximity means bringing people from different departments together into closer physical locations that enable easier and more frequent interactions between team members (Kahn and McDonough, 1997). Proximity enhances team sensemaking capability by facilitating face-to-face interactions, which are considered the richest medium of information sharing as they allow immediate feedback so that understanding can be checked and interpretations can be corrected (Koskinen et al., 2003). Proximity that allows simultaneous interaction of multiple cues, including body language, facial expression, and tone of voice, which convey knowledge beyond the spoken message, also improves team sensemaking capability by shaping and negotiating the meanings (Koskinen et al., 2003). Therefore:

H4: Team member proximity is positively related to NPD team sensemaking capability.

Open-mindedness indicates the team members' receptivity to new and possibly different ideas (Neill et al., 2007). As open-minded teams have the willingness to critically evaluate the operational routines in order to accept new ideas (Calantone et al., 2002) and revise interpretations based on new information (Maitlis and Sonenshein, 2010), they enhance their sensemaking capability. Indeed, mental models hold images of how the world works are formed by the successes and failures of the past (Jeong and Brower, 2008). As time passes, these models may no longer hold true and thus can be manipulated and changed when they represent a surrogate reality for the perceived events (Sinkula et al., 1997). Open-minded team members who are comfortable with extraordinary information and situations also notice more cues that help in anticipating environmental changes and events that facilitate sensemaking capability (Weick, 1995). Therefore:

H5: Open-mindedness is positively related to NPD team sensemaking capability.

3.2. Consequences of team sensemaking capability

Research examining the impact of sensemaking capability on information implementation (e.g., cor-

recting product-related problems) to assess the process effectiveness of NPD projects remain insufficient in the NPD literature. We argue that team sensemaking capability facilitates information implementation by providing information regarding how decisions should be carried out and by helping team members to understand the nature of the environmental change and acquire provisions for the future. Sensemaking capability also enhances information implementation by getting feedbacks from customers for the trial versions of the new products (e.g., prototypes) before the full-scale launch to uncover and correct products' problematic areas with which customers were dissatisfied.

In addition to information implementation, the impact of sensemaking capability on the speed-to-market that is the extent showing how fast a new product is developed is barely mentioned in the NPD literature. As product development process is an uncertain path through shifting markets and technologies, it requires quick and flexible options so as to cope with the unclear and changing environment (Eisenhardt and Tabrizi, 1995). Here, sensemaking capability helps team members to develop cognitive maps of their environment (Weick, 1995) by displaying shared interpretations of the environmental situations through the combination of their diverse cognitive perspectives (Bogner and Barr, 2000) and timely integration of critical information (Edmondson and Nemhard, 2009) that leads to the discovery of alternative technological solutions. Sensemaking capability also helps team members to act as a single unit to speed up product development efforts. For example, the marketing personnel contributes to the project by interpreting users' requests; in turn, the designers aid the project by drawing technical prototype details, and the engineering department assists the project by testing prototypes (Akgün et al., 2006a). Here, the higher the sensemaking capability is, the higher the interrelationships between team members would be. Therefore, it is hypothesized that:

H6: Team sensemaking capability is positively related to (1) information implementation and (2) speed-to-market.

New product success refers to the extent to which a project meets its expected commercial objectives (Cooper and Kleinschmidt, 1987). Many scholars discuss the critical factors that have positive influence on new product success, such as interpersonal conditions for teamwork, speed-to-market, and information implementation (Eisenhardt and Tabrizi, 1995; Edmondson and Nemhard, 2009). Based on the study of Akgün et al. (2006a), new product

success depends on the process effectiveness ingredients, namely speed-to-market and information implementation, such that process effectiveness mediates the relationship between the sensemaking capability and product effectiveness. Indeed, sensemaking capability does not guarantee the commercial success of a new product. New product success depends on how team members proficiently solve the customer- and product-related issues and whether the new product enters the market on time (Akgün et al., 2006a). For instance, NPD teams can only reach the best fit solutions according to the constraints and conditions of a specific environment in which there is no single definition and solution for every strategic situation by revising prototypes according to customer preferences (Neill et al., 2007). As customers not only demand higher levels of quality in new products but also expect the latest innovations, translating customer requirements into product specifications is a prerequisite for a new product success. Also, rapid product development is usually more productive and lowers cost, as lengthy time in product development tends to waste resources on peripheral activities, changes, and mistakes (Eisenhardt and Tabrizi, 1995). Therefore, it is hypothesized that:

H7: Process effectiveness, such as (1) information implementation and (2) speed-to-market, mediates the relationship between the team sensemaking capability and product effectiveness.

3.3. Mediating effect of sensemaking capability between team processes and process effectiveness

Finally, in NPD literature, there are also pieces of evidence that team process variables affect the process effectiveness variables. Zirger and Hartley (1996), for instance, reveal that 'Face-to-face communication that occurs when team members are located together helps to accelerate development by increasing mutual understanding of constraints, limitations and potential problems'. Kessler and Chakrabarti (1996) suggest that without team autonomy, too many formal reviews and approval processes inevitably result in a longer product development time. However, as we mentioned earlier in H6, sensemaking capability also impacts the process effectiveness. Taking into account these contributions, we propose that sensemaking capability mediates the relationship between the antecedent variables and process effectiveness. Indeed, if there is not any high-level communication among team members, which is one of the main dimensions of sensemaking capability, antecedents like trust, autonomy, or open-mindedness

may not yield speed-to-market or information implementation. For instance, according to Garcia et al. (2008), trust represents a necessary basis to attain a working environment with open communication. In parallel, familiarity of team members does not necessarily shorten the cycle time, as familiar team members may tend to become inward focused and neglect external communication (Brown and Eisenhardt, 1995). The vanishing of external communication restricts the information and resources from outside the team that leads the team to come up with quick solutions. Further, although open-mindedness and trust promote various considerations of solutions by offering psychological safety to the team members, a consensus on decisions can be captured through generating shared mental models. In addition, team members who share similar mental models can anticipate each other's responses and coordinate effectively when time is of the essence and opportunities for overt communication and debate are limited (Lim and Klein, 2006). Therefore:

H8: Team sensemaking capability mediates the relationship between team processes and process effectiveness.

4. Research method

4.1. Sampling

To test the hypotheses, multi-item scales adopted from prior studies for the measurement of constructs were used. All constructs were measured using five-point Likert scales ranging from 'strongly disagree' (1) to 'strongly agree' (5). By using the parallel translation method, items were first translated into Turkish by one person and then retranslated into English by a second person. The two translators then jointly reconciled all the differences. The suitability of the Turkish version of the questionnaires was then pre-tested by 10 part-time graduate students working in industry and involved in at least one NPD project. After refining the questionnaire, based on interviews with the pretest subjects, the questionnaires were distributed and collected by one of the authors, applying the 'personally administrated questionnaire' method.

The initial sample consisted of 180 firms in the industrial zone of Turkey, near Istanbul. These firms were selected because they develop new products and export them to other countries, such as the European Union, Arabic countries, Central Asia, and Russia, as identified by the Istanbul Chamber of Industry. In addition, these firms are organized and managed based on the Western management style; that is, they

operate in accordance with International Standardization Organization and European quality standards.

First, the firms' managers were contacted by telephone and the aim of our study was explained to them. Among the 180 firms contacted, we received 92 usable returns concerning NPD projects. During the sampling, to assess the product performance more accurately, we set the condition that the products used in this study must have been commercialized and launched into the marketplace for at least six months (Lynn et al., 2000). Also, the team size was an important constraint such that it should contain more than three members. Furthermore, to reduce the possible problems associated with single sourcing, we targeted the product or project manager to complete the survey. These managers were chosen because (1) product/project managers are likely to have a 'bigger-picture view' of NPD projects than other team members; (2) project managers have a broader view of each member's behaviors than other team members; and (3) it is possible that some team members in the study had preconceived opinions and expectations regarding differences in work behaviors for other team members, such as stereotypes; thus, product/project managers were expected to provide more reliable and objective data. After the respondents were selected, each respondent was informed that his or her responses would remain anonymous and would not be linked to them individually, or to their companies, or products. This was done to assure anonymity, increasing the motivation of informants to cooperate without the fear of potential reprisals. The respondents were assured that there were no right or wrong answers and that they could answer questions as honestly and forthrightly as possible. Further, a cover story was developed to prevent the respondents from filling out the questionnaire as if the measurement of the predictor variable is connected with or related to the measures of the criterion variable (Podsakoff et al., 2003).

Several industries were included, namely telecommunication, software, financial services, manufacturing and machinery, chemical, food, and material. The products were software (44.7%), consumer services (25.3%), and industrial materials, parts, and services (29.8%).

4.2. Measures

For the new product success, items developed by Cooper and Kleinschmidt (1987) were used. Six questions were asked to assess new product success with a view to assessing whether teams were meeting or exceeding profit and sales expectations. This operationalization was similar to Hackman's (1987)

view of performance operationalization as ‘meeting or exceeding the performance standards of the people who receive and/or review the team’s output’.

To measure the speed-to-market – the ability of a team to develop and launch a new product rapidly – four questions were asked. Because a multicompany and multi-industry sample was used, control for speed-to-market differences was achieved by using relative speed measures using an approach and item content similar to that of Kessler and Chakrabarti (1996). Speed-to-market was assessed relative to preset schedules, company standards, and similar competitive projects.

To assess the information implementation, we asked five questions incorporating the lessons learned during the project, and correcting problems encountered during the whole project as adapted from Lynn et al. (2000).

Regarding the team processes, we adapted Kana-wattanachai and Yoo’s (2002) question items to assess interpersonal trust. Six questions were asked to determine the level of trust. For the team autonomy, we asked four question items adapted from Sethi (2000). We asked three question items to assess the team member familiarity developed by DeChurch and Marks (2001) and Haon et al. (2009). Team member proximity constructs were adapted from Kahn and McDonough (1997) by asking two items. Regarding the open-mindedness construct, we asked four items adapted from Neill et al. (2007).

Team sensemaking capability has five components: internal communication was measured by seven items adapted from Neill et al. (2007) and Park et al. (2009). External communication was assessed by four question items adapted from Chang and Cho (2008). Information gathering, which is parallel to information acquisition in the literature, is measured by five items adapted from Moorman (1995). Information classification was assessed by asking five question items adapted from Akgün et al. (2006b). Building shared mental models’ measures are modified from Lynn et al. (2000) by asking five question items. Finally, the experimental action measures were derived from Bogner and Barr (2000) by asking five question items.

4.3. Measure validity and reliability

To assess the construct validity of the measurement instruments, a similar procedure to that of Kleijnen et al. (2007) was performed using reflective indicators for all constructs (see Appendix A). A null model with no structural relationships was estimated, and then its reliability was evaluated by means of composite scale reliability (CR) and average variance extracted (AVE)

(Fornell and Larcker, 1981; Chin, 1998). For all measures, the partial least squares (PLS)-based CR was well above the cutoff value of 0.70, and AVE values were well beyond or close to the threshold levels suggested by Fornell and Larcker (1981). In addition, the convergent validity was evaluated by inspecting the standardized loadings of the measures on their respective constructs (Chin, 1998), and all measures were found to exhibit standardized loadings that exceed 0.40 (see Appendix A). Next, as a check for discriminant validity, as suggested by Fornell and Larcker (1981), the square root of AVE for each construct is greater than the latent factor correlations between pairs of constructs (see Table 1). Moreover, as suggested by Chin (1998) and Kleijnen et al. (2007), the theta matrix (γ) was inspected, and no item was found to cross-load higher on another construct than it did on its associated construct. In addition, the correlations of the residual terms across blocks did not exceed 0.21 (Kleijnen et al., 2007). Consequently, it was determined that all constructs exhibit satisfactory discriminant validity.

4.4. Hypothesis testing

We used PLS technique to test our model. The choice was motivated by several considerations. First, according to Fornell and Bookstein (1982), PLS avoids many of the restrictive assumptions underlying maximum likelihood techniques and ensures against improper solutions and factor indeterminacy. Second, PLS is insensitive to sample size considerations and handles both very small and very large samples with more ease than does structural equation modeling (SEM) (Hair et al., 2006). Chin and Newsted (1999, p. 314), for example, wrote that ‘(For PLS) Power analysis based on the portion of the model with the largest number of predictors. Minimal recommendations range from 30 to 100 cases. (For Covariance based SEM) Ideally based on power analysis of specific model – minimal recommendations range from 200 to 800’. Finally, PLS handles both reflective and formative constructs (Hair et al., 2006).

PLS-Graph 3.0, which allows for explicit estimation of latent variable scores, and the bootstrapping resampling method were used to test the proposed model (Chin, 1998). This procedure entailed generating 500 subsamples of cases randomly selected, with replacement from the original data. Path coefficients were then generated for each randomly selected subsample. T-statistics were calculated for all coefficients, based on their stability across the subsamples, indicating which links were statistically significant.

Team sensemaking capability in product development

Table 1. Descriptive statistics

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1 Team autonomy	(0.77)													
2 Interpersonal trust	0.44***	(0.77)												
3 Team member familiarity	0.10	0.34***	(0.90)											
4 Team member proximity	-0.06	-0.03	0.07	(0.92)										
5 Open-mindedness	0.48***	0.57***	0.33***	0.01	(0.78)									
6 Internal communication	0.46***	0.62***	0.26***	0.04	0.62***	(0.67)								
7 External communication	0.35***	0.26***	0.18**	0.08	0.24***	0.46***	(0.69)							
8 Information gathering	0.37***	0.31***	0.24***	0.23***	0.36***	0.37***	0.41***	(0.72)						
9 Information classification	0.34***	0.36***	0.20**	0.14*	0.38***	0.42***	0.37***	0.49***	(0.79)					
10 Shared mental model	0.48***	0.59***	0.35***	0.02	0.50***	0.46***	0.38***	0.45***	0.51***	(0.79)				
11 Experimental action	0.27***	0.31***	0.06	-0.02	0.26***	0.23***	0.29***	0.31***	0.35***	0.45***	(0.84)			
12 Information implementation	0.47***	0.50***	0.30***	0.09	0.50***	0.48***	0.35***	0.56***	0.45***	0.59***	0.34***	(0.75)		
13 Speed-to-market	0.41***	0.37***	0.22***	0.26***	0.40***	0.43***	0.32***	0.49***	0.39***	0.45***	0.31***	0.51***	(0.81)	
14 New product success	0.46***	0.35***	0.26***	0.12	0.31***	0.39***	0.40***	0.46***	0.36***	0.49***	0.27***	0.58***	0.59***	(0.88)
Mean	3.68	4.13	3.73	2.74	3.96	3.88	3.39	3.46	3.43	3.82	3.64	3.79	3.43	3.65
Standard deviation	0.81	0.64	1.13	1.24	0.74	0.61	0.76	0.78	0.82	0.69	0.90	0.66	0.79	0.82
Skewness	-0.60	-0.93	-0.66	0.24	-0.77	-0.76	-0.34	0.09	-0.44	-0.45	-0.81	-0.01	-0.16	-0.52
Kurtosis	0.62	2.77	-0.56	-0.95	1.14	0.71	0.06	-0.15	0.39	0.79	0.66	-0.26	-0.22	0.71

* $P < 0.1$, ** $P < 0.05$, *** $P < 0.01$.

Notes: Numbers on diagonals indicate square root of average variance extracted (AVE). No correlation is greater than the corresponding square root of AVE.

Table 2. The results

Hypothesis	Relationship	Path coefficient	Results
1	Team autonomy → team sensemaking capability	0.40***	Supported
2	Interpersonal trust → team sensemaking capability	0.39***	Supported
3	Team member familiarity → team sensemaking capability	0.13	Not supported
4	Team member proximity → team sensemaking capability	0.08	Not supported
5	Open-mindedness → team sensemaking capability	0.09**	Supported
6a	Team sensemaking cap. → information implementation	0.68***	Supported
6b	Team sensemaking capability → speed-to-market	0.54***	Supported
Fit Measures	Endogenous construct		Final model
R ²	Team sensemaking capability		0.53
	Information implementation		0.46
	Speed-to-market		0.29

* $P < 0.1$, ** $P < 0.05$, *** $P < 0.01$.

During the analysis, we used the composite scores of all variables. Also, as a second-order construct, team sensemaking capability was assessed as a formative construct composed of internal and external communication, information gathering, information classification, building shared mental models, and taking experimental actions.

As shown in Table 2, the results illustrate that the hypotheses regarding the antecedents and consequences are largely confirmed. With regard to antecedents, it was found that team autonomy ($\beta = 0.40$, $P < 0.01$), interpersonal trust ($\beta = 0.39$, $P < 0.01$), and open-mindedness ($\beta = 0.09$, $P < 0.05$) have significant statistical association with team sensemaking capability, supporting H1, H2, and H5. However, any significant statistical association between team member familiarity ($\beta = 0.13$, $P > 0.1$) and team sensemaking capability was not found, thus not supporting H3. Also, team member proximity ($\beta = 0.08$, $P > 0.1$) is not significantly associated with team sensemaking capability, not supporting H4. Regarding H6, the role of team sensemaking capability on the process effectiveness, it was found that team sensemaking capability is positively related to information implementation ($\beta = 0.68$, $P < 0.01$) and speed-to-market ($\beta = 0.54$, $P < 0.01$), supporting H6.

In order to test the mediating effect of process effectiveness between sensemaking capability and new product success, including H7, we employed the Baron and Kenny (1986) procedure, where a variable (M) mediates the relationship between an independent variable (X) and a dependent variable (Y) if (a) X is significantly related to Y ; (b) X is significantly related to M ; (c) after X is controlled for, M remains significantly related to Y ; and (d) after M is controlled for, the X – Y relationship is zero. Steps (b) and (c) are the essential steps in establishing mediation and Step

(d) is only necessary to prove a fully mediated effect. Also, the presence of the mediator (M) must reduce the impact of the independent variable on the outcome compared with when M is not present. Further, entering the mediator into the PLS-based SEM model should also result in a significant increase in R^2 . In this way, we performed three different PLS-based SEM models as shown in Table 3:

- Model a, including team sensemaking capability construct (X) and the new product success (Y), indicates that team sensemaking capability ($\beta = 0.52$, $P < 0.01$) is positively related to the new product success, and $R^2_{\text{NPS}} = 0.27$.
- Model b, covering the team sensemaking capability construct (X) and process effectiveness (M) variables, shows that team sensemaking capability is positively associated with information implementation ($\beta = 0.69$, $P < 0.01$) and speed-to-market ($\beta = 0.54$, $P < 0.01$), and $R^2_{\text{inf. imp.}} = 0.47$, $R^2_{\text{speed}} = 0.30$.
- After team sensemaking capability construct (X) is controlled, as shown in model c, it was found that information implementation (M) ($\beta = 0.68$, $P < 0.01$) and speed-to-market (M) ($\beta = 0.54$, $P < 0.01$) are positively associated with new product success (Y). Also, process effectiveness variables reduce the effect of team sensemaking capability on the new product success, and inclusion of the information implementation and speed-to-market in the model increases the R^2 of new product success ($R^2_{\text{NPS}} = 0.47$).

Based on the previous results, process effectiveness variables, information implementation, and speed-to-market mediate the relationship between team sensemaking capability and new product success, supporting H7.

Team sensemaking capability in product development

Table 3. Results of mediating role of process effectiveness

Relationship	Model a	Model b	Model c
Team sensemaking capability → information implementation		0.69***	0.68***
Team sensemaking capability → speed-to-market		0.54***	0.54***
Information implementation → new product success			0.36***
Speed-to-market → new product success			0.35***
Team sensemaking capability → new product success	0.52***		0.06
	$R^2_{\text{NPS}} = 0.27$	$R^2_{\text{inf.imp}} = 0.47$	$R^2_{\text{NPS}} = 0.47$
		$R^2_{\text{speed}} = 0.30$	$R^2_{\text{inf.imp}} = 0.46$
			$R^2_{\text{speed}} = 0.30$

* $P < 0.1$, ** $P < 0.05$, *** $P < 0.01$.

Table 4. Results of mediating role of team sensemaking capability

Relationship	Model a	Model b	Model c
Team autonomy → information implementation	0.34***		0.16
Interpersonal trust → information implementation	0.41***		0.22
Team member familiarity → information implementation	0.03		−0.05
Team member proximity → information implementation	0.15		0.10
Open-mindedness → information implementation	0.07		0.03
Team autonomy → speed-to-market	0.34***		0.20*
Interpersonal trust → speed-to-market	0.20		0.08
Team member familiarity → speed-to-market	0.07		0.02
Team member proximity → speed-to-market	0.26**		0.23**
Open-mindedness → speed-to-market	0.19		−0.16
Team autonomy → team sensemaking capability		0.40***	0.40***
Interpersonal trust → team sensemaking cap.		0.39***	0.39***
Team member familiarity → team sensemaking cap.		0.13	0.13
Team member proximity → team sensemaking cap.		0.09	0.09
Open-mindedness → team sensemaking capability		0.08**	0.08
Team sensemaking capability → information implementation			0.46***
Team sensemaking capability → speed-to-market			0.33***
	$R^2_{\text{inf.imp}} = 0.45$	$R^2_{\text{SMC}} = 0.57$	$R^2_{\text{SMC}} = 0.57$
	$R^2_{\text{speed}} = 0.35$		$R^2_{\text{inf.imp}} = 0.53$
			$R^2_{\text{speed}} = 0.39$

* $P < 0.1$, ** $P < 0.05$, *** $P < 0.01$.

Also, we employed the Baron and Kenny (1986) procedures to examine the mediating effect of team sensemaking capability between team process variables and process effectiveness (H8). As shown in Table 4:

- Model a, including team process variables (X) and the process effectiveness (Y), indicates that team autonomy ($\beta = 0.34$, $P < 0.01$) and interpersonal trust ($\beta = 0.41$, $P < 0.01$) are positively related to information implementation; and team autonomy ($\beta = 0.34$, $P < 0.01$) and team member proximity

($\beta = 0.26$, $P < 0.05$) are positively associated with speed-to-market; and $R^2_{\text{inf.imp.}} = 0.45$, $R^2_{\text{speed}} = 0.35$.

- Model b, covering the team process variables (X) and team sensemaking capability (M) construct, shows that team autonomy ($\beta = 0.40$, $P < 0.01$), interpersonal trust ($\beta = 0.39$, $P < 0.01$), and open-mindedness ($\beta = 0.08$, $P < 0.05$) are positively associated with team sensemaking capability, and $R^2_{\text{SMC}} = 0.57$.
- After team process variables (X) are controlled, as shown in model c, it was found that team

sensemaking capability (M) is positively associated with information implementation ($\beta = 0.46$, $P < 0.01$) and speed-to-market ($\beta = 0.33$, $P < 0.01$). However, some antecedent variables, such as team autonomy and team member proximity are still positively related to speed-to-market. Also, team sensemaking capability reduces the effects of interpersonal relationships and team process variables on the process effectiveness, and inclusion of the team sensemaking capability in the model increases the R^2 of information implementation and speed-to-market, such as $R^2_{\text{inf. imp.}} = 0.53$, $R^2_{\text{speed}} = 0.39$.

Based on the previous results, sensemaking capability mediates the relationship between team processes and information implementation, while it partially mediates the relationship between team processes and speed-to-market, partially supporting H8.

4.5. Discussion and implications

This study empirically showed that team sensemaking capability is important to enhance both information implementation and speed-to-market. While previous studies investigated the role of a variety of factors such as IT usage (Barczak et al., 2008), decision making (Dayan and Elbanna, 2011), strategic market orientation (Mu and Di Benedetto, 2011), and NPD climate (Parry et al., 2009) on the speed-to-market and process effectiveness, our study specifically demonstrated the impact of team sensemaking capability on the process effectiveness. We found that team sensemaking capability helps NPD teams to take outstanding jobs uncovering product problem areas with which customers were dissatisfied, improve knowledge implementation, and share the information quickly that leads to the detection and correction of product-related errors accurately. In addition, team sensemaking capability decreases the time spent for NPD by helping team members to organize the market and technical information in a meaningful way and by reducing project-related complexity.

This study also empirically demonstrated that when team members believe that their teammates are professional in their jobs, dedicated to their responsibilities as well as duties, respond to the problems constructively and with more care, and have the willingness to listen to the work-related problems (i.e., *interpersonal trust*), teams can (1) communicate internally and externally; (2) gather and classify the information; (3) build a shared mental model; and (4) take more effective and successful experimental

actions. These findings support the previous studies (Dirks and Ferrin, 2001; Dreu and Vianen, 2001; Muethel et al., 2012). As such, our findings are parallel with empirical studies that show that interpersonal trust enhances communication and information sharing (Dirks and Ferrin, 2001), and promotes interpersonal relationships and collaborative culture (Kyriazis et al., 2012). However, at the same time, it should be noted that our study compiled the fragmented studies on the information processing and communication variables under the umbrella of team sensemaking capability. Specifically, our study empirically showed that interpersonal trust entails individuals' beliefs in the goodwill of their teammates and confidence in their ability to complete a task and succeed, and thereby influences their voluntary and spontaneous rather than systematic and planned activities. In addition, the results indicated that interpersonal trust entails a prediction about the team members' behaviors, reducing the perceived risk and opportunism, and promoting cooperative problem-solving activity in groups (Zaheer et al., 1998). This study also showed that when top management does not interfere with the task undertaken within a team, so that team members have a major role in making important decisions about the product (i.e., *team autonomy*), teams improve their sensemaking capabilities. Our findings are consistent with studies of Van den Ende and Wijnberg (2003), who found that autonomy provides a platform for information exchange by conducting case studies of innovation teams; of Jassawalla and Sashittal (1998), who showed that autonomy enhances cross-functional collaboration; and of Donnellon (1993), who illustrated that team autonomy helps in the accomplishment of team-related work by investigating 12 product development teams. However, it should be taken into consideration that prior investigations have not produced a thorough understanding of how team autonomy affects team sensemaking capability construct. In our study, it appears that team autonomy increases team ownership, commonality, and cooperative negotiation; leads to focus on real-time planning, scheduling, and resource allocation; and opens up a space for taking initiatives regarding the reconfiguration of project-related routines to enhance sensemaking capability (Nakata and Im, 2010). This study, next, empirically illustrated that when managers encourage employees to think outside the box and develop original ideas that are highly valued in the organization (i.e., *open-mindedness*), team members can question existing assumptions, organize the technical and market-related information into meaningful ways, and develop a better vision for the project. This finding is similar to the organizational level of studies

(Neill et al., 2007). At the same time, this finding provides empirical evidence to previous theoretical studies on the team level of analysis, such that Tjosvold and Poon (1998) and Troy et al. (2001) have implicitly noted that creating and promoting open-mindedness affect sensemaking in teams. Specifically, our findings showed that open-mindedness promotes direct and respectful engagement even in the presence of opposing views, rather than avoidance in case of disagreement, and leads to goal cooperation rather than goal competition to improve a team's sensemaking capability. Also, open-mindedness helps people to actively question and discuss project-related events, and introspectively examine the origins and applicability of project-related routines (Baker and Sinkula, 1999; Farrell and Mavondo, 2004).

Our results unexpectedly showed that team member proximity and familiarity do not have a positive effect on team sensemaking capacity. It was expected that when the core engineers in the team are located so close to the core marketers, they could talk to one another without using a telephone (i.e., *proximity*); thus, the team improves its sensemaking capability. The reason might be that today's advanced information and communication technologies eliminate the necessity for high proximity of team members and allow virtual teams to even produce superior performance on innovative tasks (Hoegl and Proserpio, 2004). Also, it appears that through close proximity, people concentrate more to the local information at the expense of more remote information and place attention disproportionately at the local information rather than the team level, potentially distracting them from attending to the team as a whole – reducing sensemaking capability of teams. Next, it should be noted that, consistent with previous studies, team member proximity can be a moderator variable (Hoegl et al., 2007), or the relationship between team proximity and sensemaking capability is not always straightforward and may depend on some contextual conditions (Chong et al., 2012). Also, we found that there is no statistical association between team sensemaking capability and team member familiarity. It looks like when the team members know each other prior to the formation of the project team (i.e., *team familiarity*), the team members utilize each others' knowledge without critically evaluating their reliability (Akgün et al., 2006a). In addition, the results suggest that prior interpersonal knowledge about other group members may lead to larger bias toward discussing shared information, which is consistent with the study of Harrison et al. (2003) and Gruenfeld et al. (1996), who suggest that during group discussion, the prob-

ability that a given piece of information will be mentioned increases with the number of members who are already aware of it.

Next, this study showed that information implementation and speed-to-market mediate the relationship between the team's sensemaking capability and new product success. When team members have a better internal/external communication, information gathering and classification, build a shared mental model, and take experimental actions, they develop products that meet the volume expectations, overall sales expectations, profit expectations, return on investment expectations, and senior management expectations by the ways of uncovering and correcting product problem areas with which customers were dissatisfied and developing and launching the product faster than the major competitor or in less time than what was considered normal in the industry. While previous studies investigated the mediating role of speed-to-market/information implementation between new product success and cross-functional integration (McNally et al., 2011), team culture (Akgün et al., 2010), product quality and innovativeness (Molina-Castillo et al., 2011), and organizational design (Blazevic and Lievens, 2004), we specifically investigated the link between team sensemaking capability and new product success. Our results suggest that team sensemaking capability is actualized through actions and practices to leverage the success of new products in the market, such that new products' success is incomplete if people do not use sensemaking capability of teams envisioned in the information implementation and speed-to-market. In a sense, information implementation enhances a team's knowledge base and people's attitudes and perceptions on the project-related issues and events and symbolizes the commitment and dedication of organizational actors to rational behavior leveraging the link between sensemaking capability and new product performance. Also, team sensemaking capability leads to the development of new opportunities during the project and early improvement of the product development process by fastening the decision-making process, and helps the project team to develop products quickly for improving the performance of new products in the market.

Finally, this study illustrated that team sensemaking capability mediates the relationship between team processes and information implementation during the projects, such that the team's sensemaking capability, through the accounts it generated based on team autonomy, shared mental models, and trust, affects information utilization to solve project and product-related problems. These findings extend prior research empirically. For instance, Bstieler

(2006) demonstrated that interpersonal trust enhances product development effectiveness. Jassawalla and Sashittal (1998) noted that open-mindedness leads to creative products and product-related problem solving. Moorman and Miner (1998) mentioned that team autonomy facilitates team learning. However, our findings in Table 4 show that interpersonal trust, autonomy, open-mindedness, etc. unfold their beneficial potential on team effectiveness by leveraging teams' sensemaking capability (Barczak et al., 2010). These results indicate that sensemaking capability makes information implementation possible by examining the interpersonal and social interactions and structures involved in the recursive processes of a product development project and by reproducing social structures and enabling and constraining interpersonal relations. In other words, interpersonal interactions, together with sensemaking, create environments that subsequently become available for creating the social reality where individuals operate to solve product-related problems and speed up the development process. It was also found that sensemaking capability provides a valve to release interaction tension and clarify, organize, and rationalize the ambiguity involved in the product development process. Here, sensemaking capability leads to a collective understanding of a situation, developing a strategic model of the intervention and defining a shared understanding of a desired outcome. Therefore, sensemaking entails not only developing a common interpretation or a shared frame of reference but also developing one or more mutual solutions or agreed responses that are selected from among the many possibilities.

From this research, the management should (1) establish a sense of self-administration in the team where needed; (2) encourage employees to think outside the box; (3) emphasize open-mindedness; (4) seed the establishment of trust among team members; (5) enhance the teamwork; and (6) promote freedom of thought and freedom of speech in the discussions regarding NPD process. Management should also facilitate the internal and the external communication to enhance the team sensemaking capability. Here, management can facilitate regular team meetings, set conditions to enable team members to share information, and bring customers and outside experts into the teams. Management should further help teams to continuously collect information from customers and competitors' activities, help team members to organize market- and technology-related information in meaningful ways, clarify the vision of the project to the team members, and enable the customers to access the prototypes prior to the launch of the product.

5. Limitations and future research

There are some methodological limitations to this study – notably, single sourcing and self-reporting. Scholars (Aviolo et al., 1991) have argued that studies employing single-source methodology can be biased by artificially high intercorrelations produced by an overall response tendency. Even though, as noted, project managers are able to see the bigger picture of project activities, by collecting data from team members as respondents in each of the NPD projects who will add new insights about these complex constructs, such as team sensemaking capability. Also, because there was a time lag between the times when the projects were completed and when data were collected, there might be a recall issue in survey questions. Next, utilizing a cross-sectional design with questionnaires was one of the limitations of this study. The generalizability of sampling is another limitation of this study. The study was conducted in a specific national context that involves only Turkish firms. It is important to note that readers should be cautious when generalizing the results to different cultural contexts. Finally, the limitation of our analysis technique is that PLS focuses much more on prediction and does not provide a test of theoretical fit (Hair et al., 2006).

The sensemaking capability of teams triggers the opportunity for future research. Here, the presented model in this study is broad and does not capture the possible moderating effects of environmental turbulence, uncertainty, and routines of the projects. In addition, team leader characteristics and IT usage may affect the relationships between team sensemaking capability and project outcomes. Also, the relationship between team sensemaking capability, and proximity and familiarity may not be linear, which warrants an empirical investigation (Uzzi and Spiro, 2005). For future researches, the determinants of team sensemaking capability can be expanded, such as team culture values (e.g., employee orientation, customer orientation, innovativeness orientation, social responsibility) and types of team memory (e.g., declarative and procedural memory) (Lynn and Akgün, 2000).

6. Conclusion

Team sensemaking capability is one of the core competencies of firms. However, how team sensemaking capability can be enhanced and its effect on the project success are missing and should be added to the TIM literature. In this study, we tested the role of team process variables on the team's sensemaking

capability, and the impact of the team sensemaking capability on the process and product effectiveness variables. Our results confirm that team sensemaking capability has a significant effect on the information implementation, speed-to-market, and new product success. Also, our conclusions demonstrate that team autonomy, interpersonal trust, and open-mindedness impact the development of the team sensemaking capability.

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2. This definition is close to the well-established concept of market orientation, which is defined as organization-wide generation, dissemination, and responsiveness to market intelligence (Jaworski and Kohli, 1990). However, we propose that sensemaking capability offers several insights into the market orientation perspective. For instance, whereas sensemaking capability perspective highlights the importance of both the internal and the external information sources, market orientation scholars tend to focus more on obtaining information from external sources (Jaworski and Kohli, 1996). Also, sensemaking literature argues the importance of building shared mental models between team members (Ashmos and Nathan, 2002) whereas market orientation literature suggests that the interpretation of market information is made by the person(s) who acquire the information prior to its dissemination (Jaworski and Kohli, 1996) and, therefore, does not emphasize the construction of collective mind. Further, although sensemaking capability is driven by plausibility rather than accuracy (Seligman, 2006) and therefore gives the opportunity to deal with uncertainty (Weick et al., 2005), market orientation literature focuses on collecting the existing information about customers and competitors.
3. Scholars (e.g., Daft and Weick, 1984; Thomas et al., 1993; Jeong and Brower, 2008) note that sensemaking as a series of cognitive processes is usually interviewed in three steps, such as noticing or scanning, interpretation, and actioning, interconnected through a feedback loop in the literature. Noticing is concerned with data collection (Daft and Weick, 1984). NPD teams are often formed for the purpose of collecting necessary information about both external and internal environments (Ashmos and Nathan, 2002). Interpretation involves the development or application of ways of comprehending the meaning of information. It entails the fitting of information into some structure for understanding and action (Thomas et al., 1993). Interpretation arises through the communication between team members and leads to actions which means new responses based on the interpretation (Daft and Weick, 1984). The outcome of action often provides a pretext that leads to new ways of noticing and interpretation (Jeong and Brower, 2008).

Notes

1. From a behaviorist perspective, team learning is considered to be incorporating lessons learned during the project and as correcting problems encountered, operationalized as information/knowledge implementation or usage by Moorman (1995) and Lynn et al. (2000). Sarin and McDermott (2003) also pointed out that team learning involves making use of information, which leads to the detection and correction of errors and improves the likelihood of effective NPD. As this aspect of team learning, information implementation, has been used widely in NPD literature and affords us a measurable outcome, we adapted this definition for this study.

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Appendix A. Measures

CR: Composite reliability

AVE: Average variance extracted

Factor loading are shown in parenthesis

Antecedents

Interpersonal trust

- Most of my teammates approach his/her job with professionalism and dedication. (0.76)
- I see no reason to doubt about my teammates' competence and preparation for the job. (0.77)
- Most of my teammates can be relied on what they will do. (0.82)
- I can talk freely to my team about difficulties I am having at work and know that my team is willing to listen. (0.72)
- If I share my problems with my team, I know he/she would respond constructively and caringly. (0.83)
- I would have to say that we (my team) have made considerable emotional investments in our working relationship. (0.71)

CR = 0.90

AVE = 0.60

Team sensemaking capability in product development

Team autonomy

- Unless help wanted, top management did not interfere with the teams' job. (0.65)
- Top management provided the self-administration the team needed. (0.82)
- The team had a major role in making important decisions about the product (0.77)
- The team was allowed to do the project work as it deemed fit. (0.83)

CR = 0.85

AVE = 0.59

Team member proximity

- The core engineers on this team were located within a short walk of the core marketers. (0.92)
- The core engineers on this team were located so close to the core marketers that they could talk to one another without using a telephone. (0.92)

CR = 0.92

AVE = 0.85

Team member familiarity

- I knew the other members of my team (on average) at the time our project team was formed. (0.93)
- I had interaction with the other members of my team (on average) at the time our project team was formed. (0.95)
- Members having participated in the project were well acquainted with the skills of the others. (0.81)

CR = 0.93

AVE = 0.81

Open-mindedness

- Our team places a high value on open-mindedness. (0.78)
- Managers encourage employees to think outside the box. (0.85)
- Original ideas are highly valued in our team (0.82)
- We are not afraid to reflect critically on the shared assumptions we have about the way we do anything during the projects. (0.64)

CR = 0.86

AVE = 0.61

Team sensemaking capacity

Internal communication

- Our team has regular interdepartmental meetings to discuss market trends and developments. (0.65)

- Team members shared the information among the team members. (0.71)
- Team members had lively communication among the team members. (0.86)
- The information our team members shared was useful. (0.70)
- Team members conducted frequent informal communications at lunch or after work with fellow project team members. (0.69)
- Team members conducted frequent formal communications through team meetings with fellow project team members. (0.66)
- Team members educated each other during the project. (0.71)

CR = 0.84

AVE = 0.45

External communication

- Our team regularly discussed with customers. (0.62)
- Our team systematically acquired external technological and market information. (0.82)
- Our team consulted regularly to outside experts. (0.65)
- Our team conducted systematic benchmarking. (0.71)

CR = 0.77

AVE = 0.47

Information gathering

- The team had the ability of continuously collecting information from customers. (0.72)
- The team had the ability of continuously collecting information about competitors' activities. (0.71)
- The team had the ability of continuously collecting information about relevant publics other than customers and competitors. (0.76)
- The team had the ability of continuously reexamining the value of information collected in previous studies. (0.76)
- The team had the ability of continuously collecting information from external experts such as consultants. (0.62)

CR = 0.84

AVE = 0.52

Information Classification

- Information collected by the team (for example, test results) was coded and sorted to be understood easily by other team members. (0.77)

- Market information was organized in meaningful ways. (0.75)
- Technical information was organized in meaningful ways. (0.85)
- Technical information was summarized to reduce its complexity. (0.87)
- Market information was summarized to reduce its complexity. (0.69)

CR = 0.89

AVE = 0.63

Building shared mental models

- Team members developed a common language during the project. (0.70)
- The team had a shared understanding of the target market (user). (0.81)
- The team had a shared understanding of target customers' needs and wants. (0.84)
- The team had a shared understanding of the required product features. (0.85)
- Our team members had a shared vision during the project. (0.77)

CR = 0.89

AVE = 0.63

Taking experimental actions

- Our team lets the customers meet the prototypes before launching the product. (0.81)
- Our team does not complete the design stage before the customers meet the prototypes. (0.80)
- Our top management highly encourages the customers to meet the prototypes prelaunch. (0.81)
- Our team finishes the design of the product only after taking the feedback of the customers. (0.89)
- Feedback of the customers after meeting the prototypes offers key solutions to our team. (0.88)

CR = 0.92

AVE = 0.71

Consequences

New product success

Our product:

- Met or exceeded volume expectations. (0.84)
- Met or exceeded the first year number expected to be produced and commercialized. (0.89)
- Met or exceeded overall sales expectations. (0.90)
- Met or exceeded profit expectations. (0.91)
- Met or exceeded return on investment expectations. (0.90)
- Met or exceeded senior management expectations. (0.86)

Team sensemaking capability in product development

CR = 0.95

AVE = 0.78

Speed-to-market

This product:

- Was developed and launched (fielded) faster than the major competitor for a similar product. (0.76)
- Was completed in less time than what was considered normal and customary for our industry. (0.85)
- Was launched on or ahead of the original schedule developed at initial project go-ahead. (0.80)
- Pleased the top management with the time it took us from specs to full commercialization. (0.79)

CR = 0.88

AVE = 0.65

Information implementation

- Post-launch, this product had far fewer technical problems than our nearest competitor's product or our own previous products. (0.65)
- Market easily adapted to this product without having any problem. (0.77)
- Overall, the team did an outstanding job uncovering product problem areas with which customers were dissatisfied. (0.85)
- Overall, the team did an outstanding job correcting product problem areas with which customers were dissatisfied. (0.83)
- Team members shared the information immediately face-to-face, via telephone, or e-mail before any important facility. (0.66)

CR = 0.87

AVE = 0.57